

ViTGAN: Generative Adversarial Networks with Vision Transformers

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Abstract

We introduce ViTGAN, a purely transformer-based generative adversarial network for high-fidelity image synthesis. Our model removes convolutional inductive biases and demonstrates competitive results on standard benchmarks such as CIFAR-10 and CelebA. arXiv:2107.04589

1 Introduction

Generative adversarial networks (GANs) have become a dominant paradigm for image synthesis since their introduction. Recently, vision transformers have shown strong performance on discriminative tasks, motivating their use in generative settings as well.

2 Related Work

2.1 Generative Adversarial Networks

GANs consist of a generator and a discriminator trained in a minimax game. Numerous architectural improvements have been proposed to stabilize training and improve sample quality.

2.2 Vision Transformers

Vision Transformers (ViT) apply the standard transformer architecture directly to sequences of image patches, achieving strong results on image classification.

3 ViTGAN

3.1 Discriminator

The discriminator treats an image as a sequence of patches and processes them with a standard transformer encoder, followed by a classification head.

3.2 Generator

The generator maps a latent vector to a sequence of patch embeddings which are decoded into image patches and assembled into the final output image.

4 Experiments

We evaluate ViTGAN on CIFAR-10 and CelebA, reporting FID scores compared to convolutional baselines. Our model achieves competitive performance without any convolutional layers.

5 Conclusion

We presented ViTGAN, demonstrating that pure transformer architectures are viable for high-fidelity image generation, opening avenues for future convolution-free generative models.

References

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